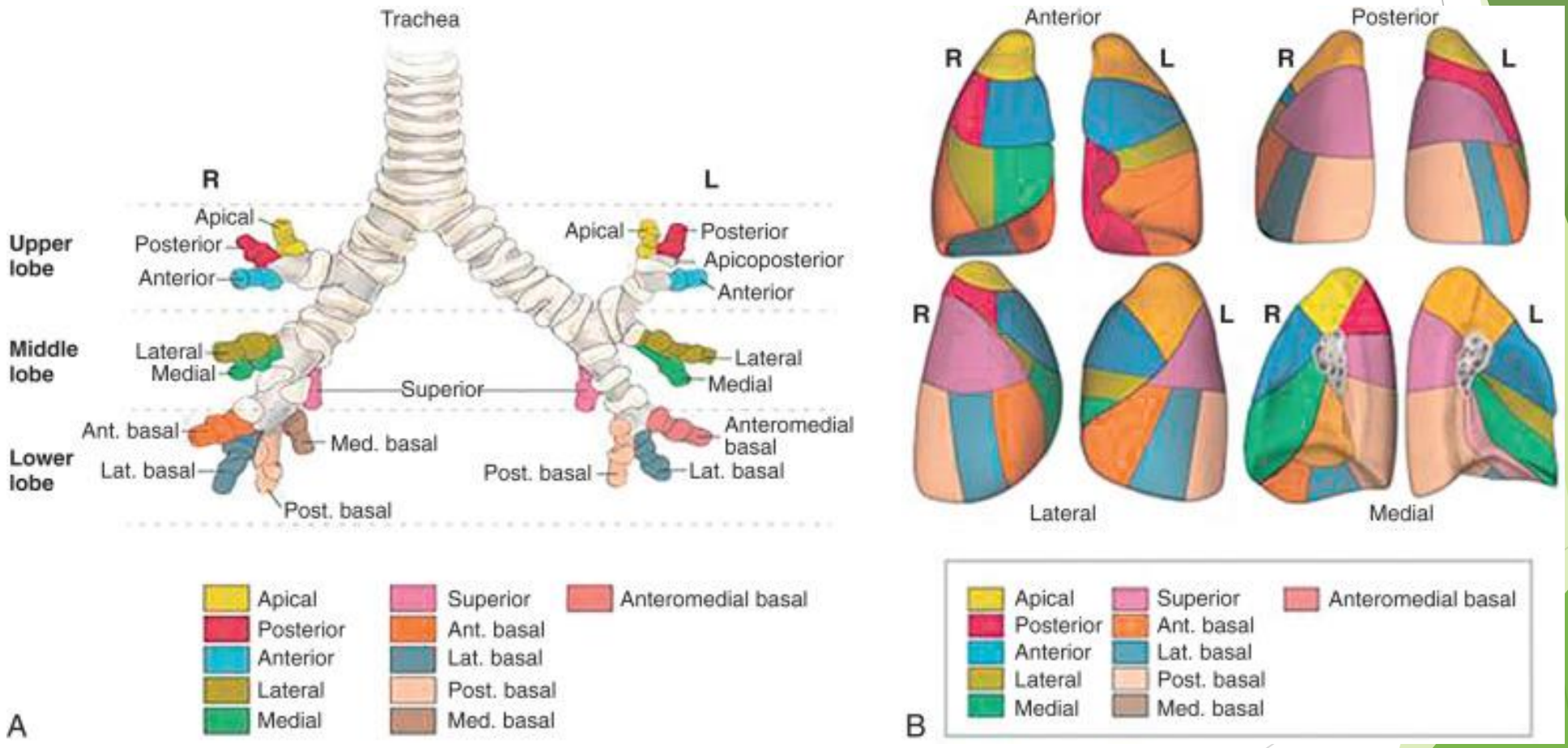
CT Lung Anatomy: A Foundation for Diagnosis Part 2

Dr Ahmed Elsawaf MD

LUNG SEGMENTS



Right Lung (10 segments)

RUL: 3 segments - **Apical, Posterior, Anterior**

RML: 2 segments - **Lateral, Medial**

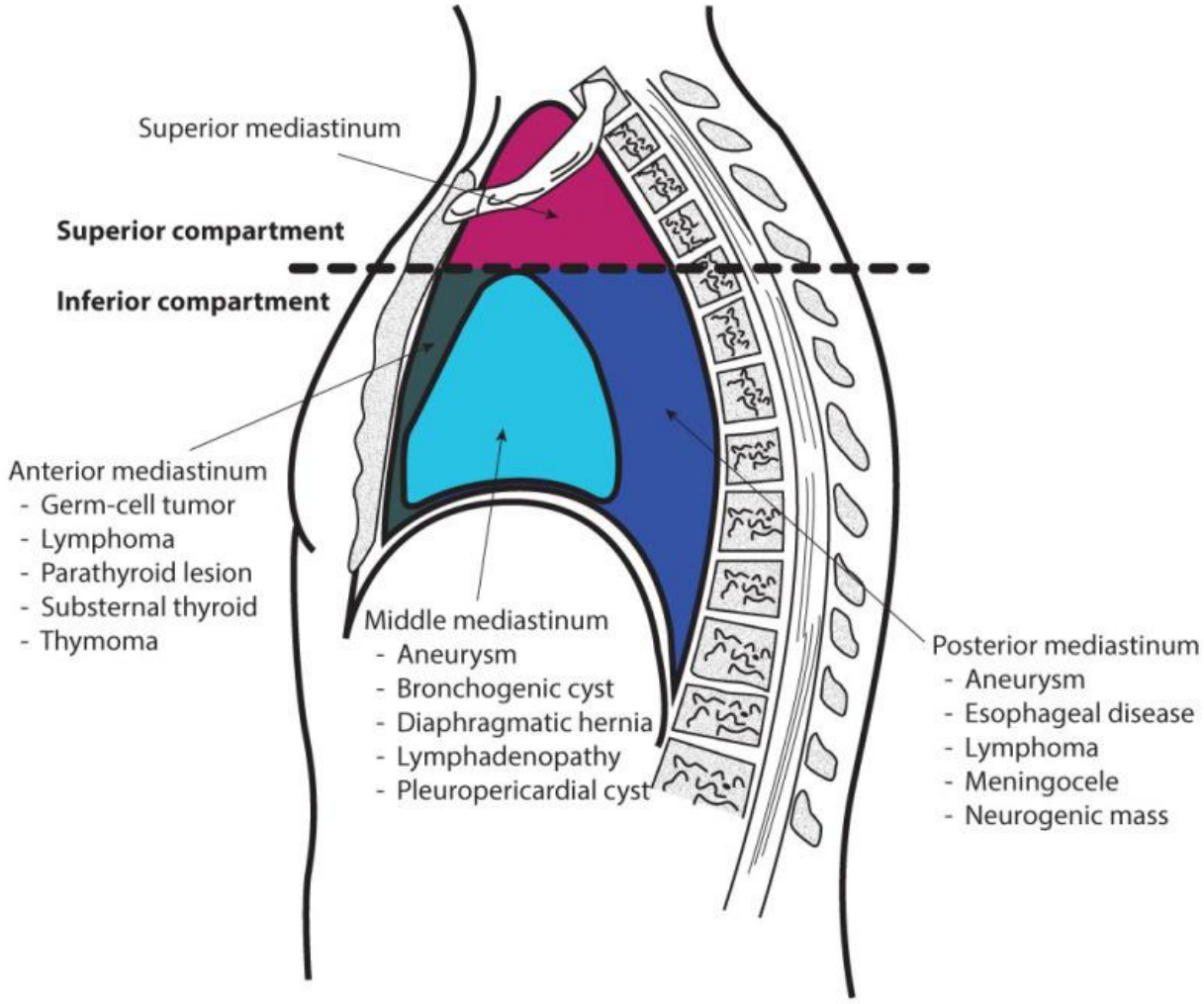
RLL: 5 segments - **Superior, Medial basal, Anterior basal, Lateral basal, Posterior basal**

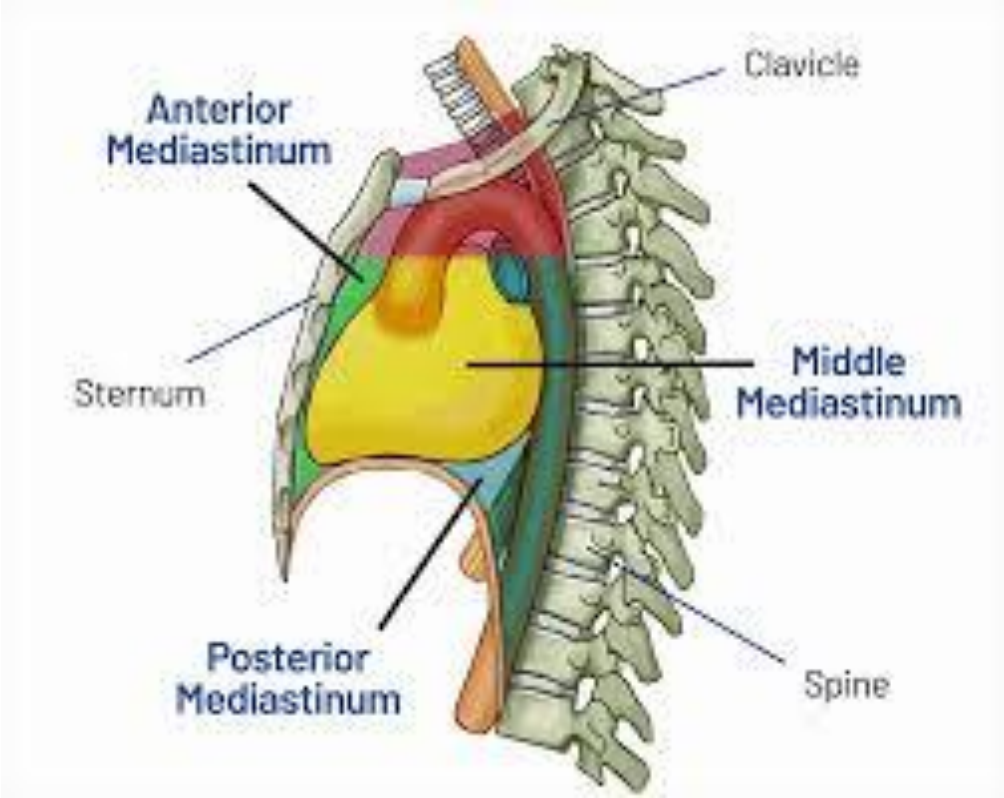
Left Lung (8 segments)

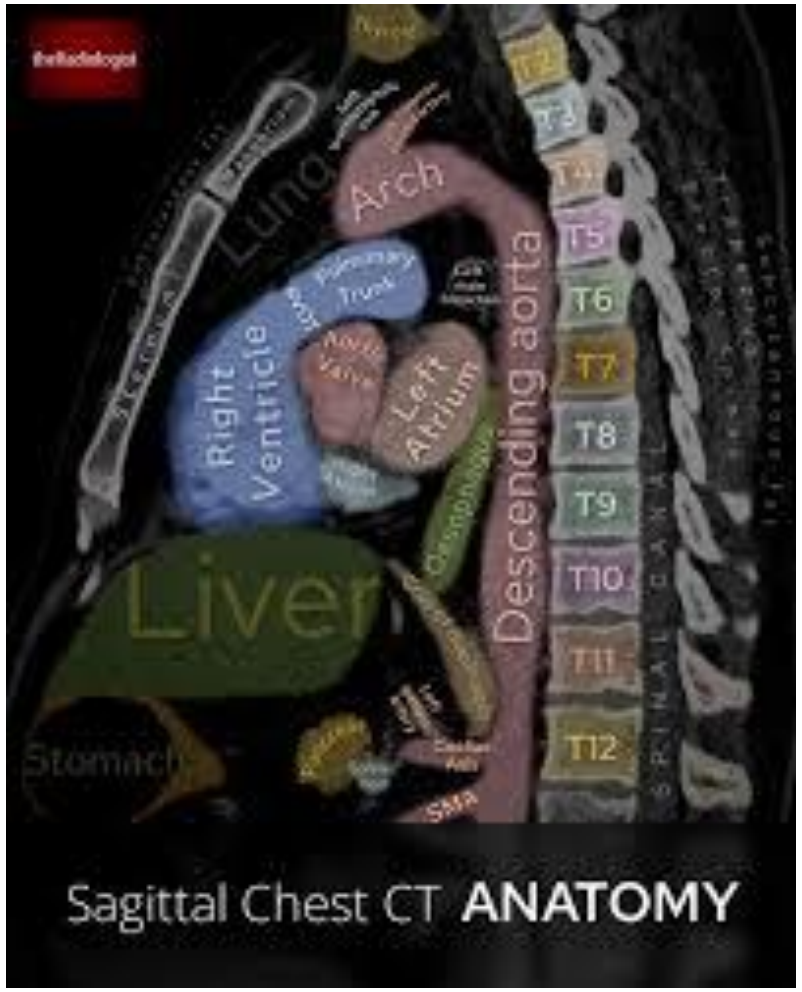
LUL: 4 segments. **The apical and posterior segments often fuse to form the apicoposterior segment**. The anterior segment. The lingula has a **superior and inferior** segment, analogous to the RML.

LLL: Usually 4 segments. The superior, anterior, lateral basal and posterior basal**……(no medial segment)**

The Mediastinum: A Compartmental Approach




Clavicle
Anterior Mediastinum
Sternum
Middle Mediastinum
Posterior Mediastinum
Spine


Lung
Arch
Right Ventricle
Left Atrium
Descending aorta
Liver
Stomach
SMA
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
Sagittal Chest CT ANATOMY

Use the simplified 4-compartment model:

Superior Mediastinum: aortic arch and branches (brachiocephalic, left common carotid and left subclavian), two brachiocephalic veins (ending into SVC)

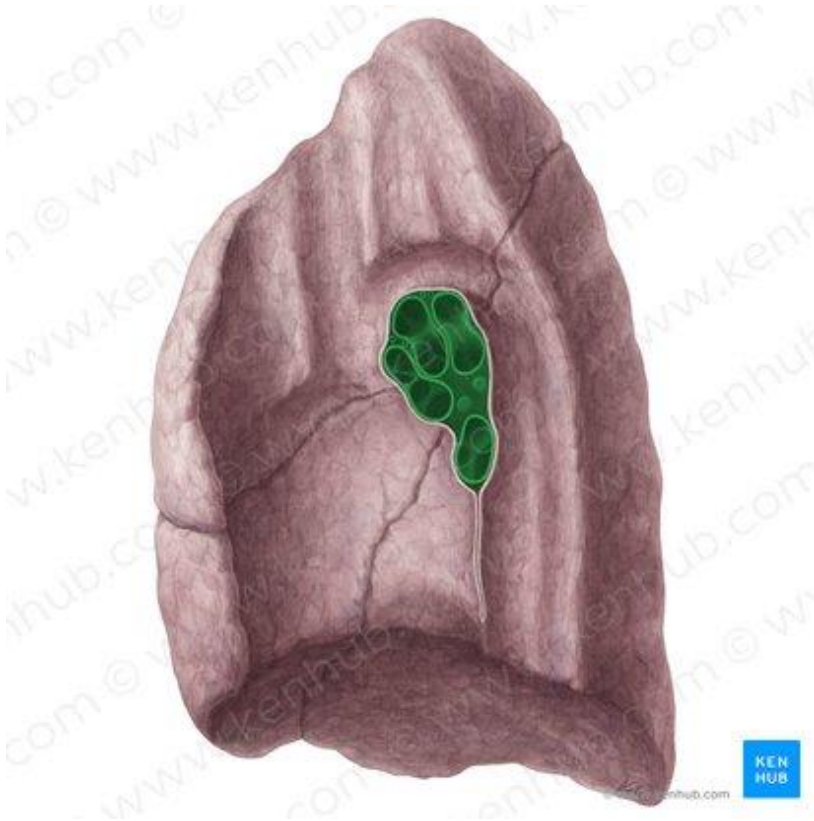
Anterior Mediastinum: Thymus, lymph nodes, fat.

Middle Mediastinum: Heart, pericardium, great vessels

Posterior Mediastinum: Descending aorta, esophagus, azygos/hemiazygos veins, nerves, lymph nodes.

- **Clinical Pearl:** "When you see a mass, first localize it to a compartment. This instantly narrows your differential diagnosis from 50 possibilities to about 5."

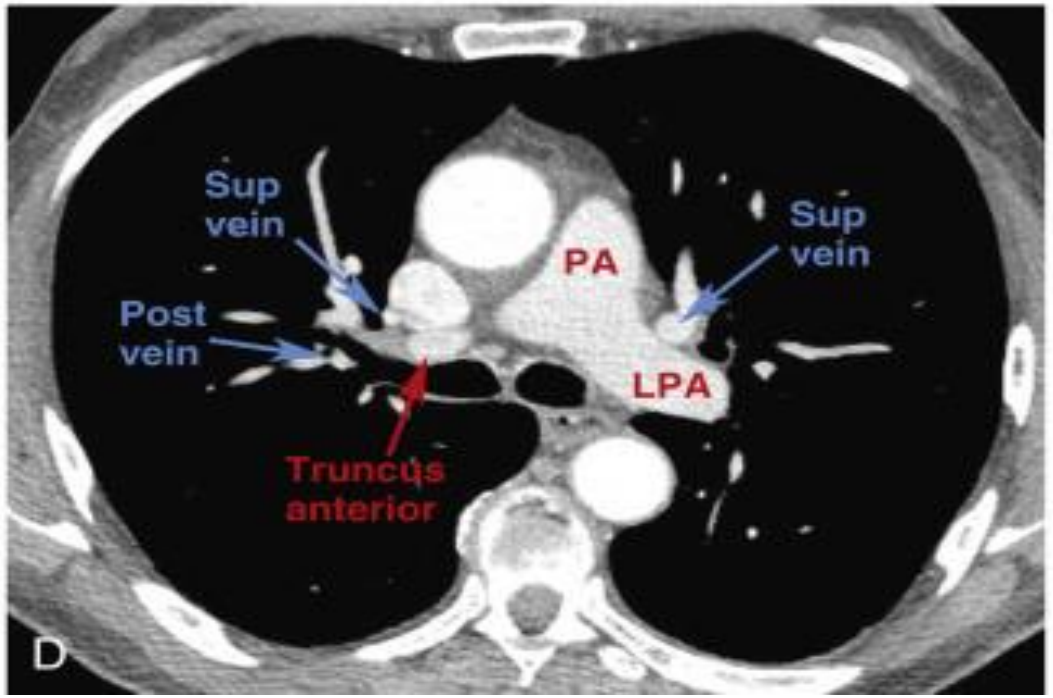
The Hilum: Don't Get Lost



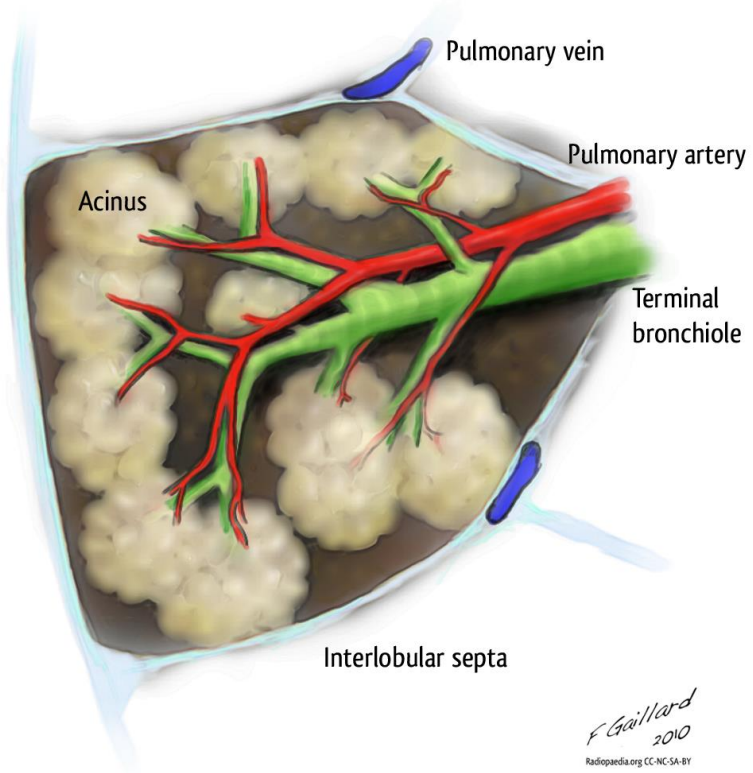
The Hilum: Don't Get Lost

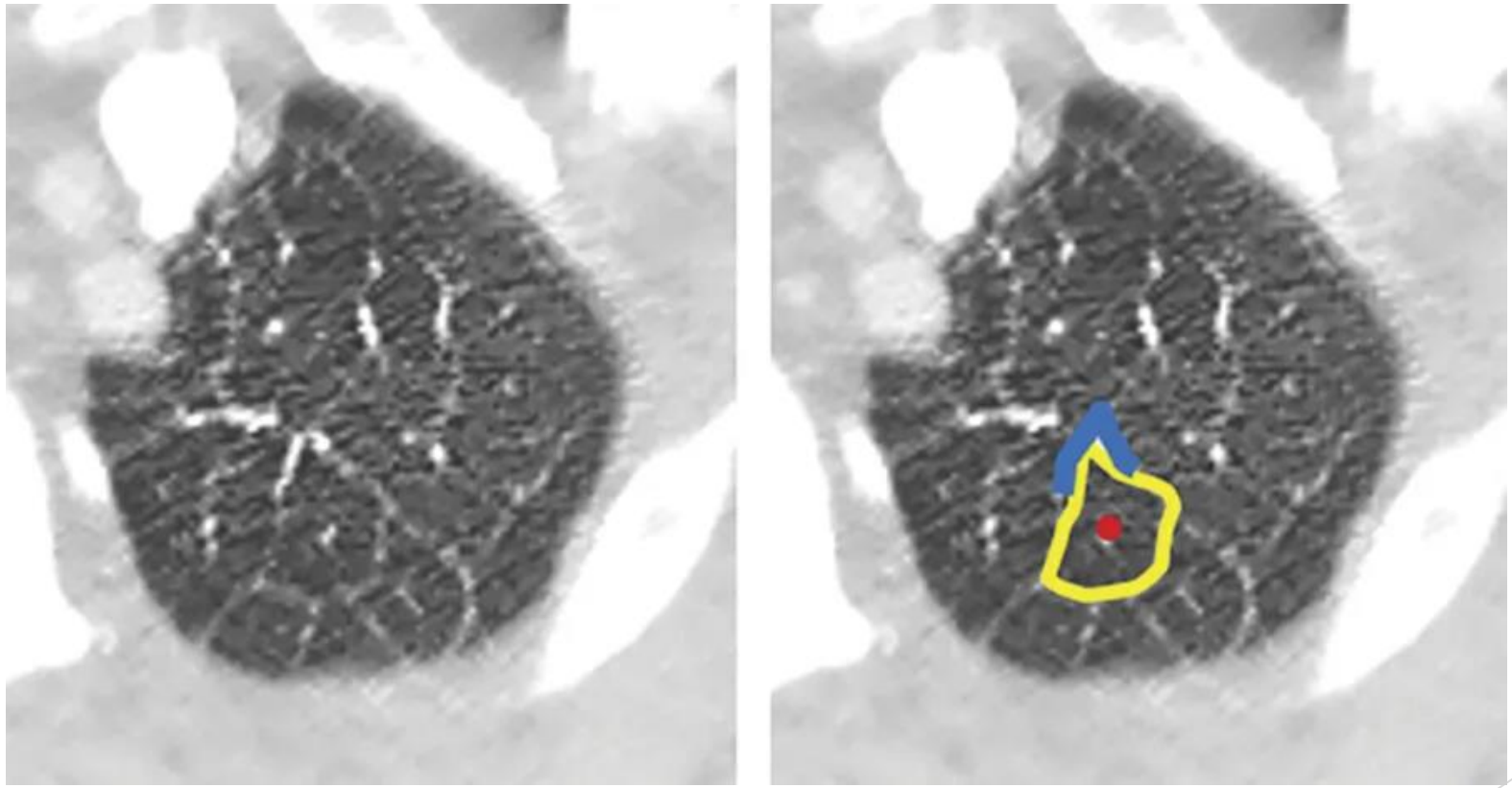
- From anterior to posterior:
- **Vein, Artery, Bronchus**
- (V-A-B).

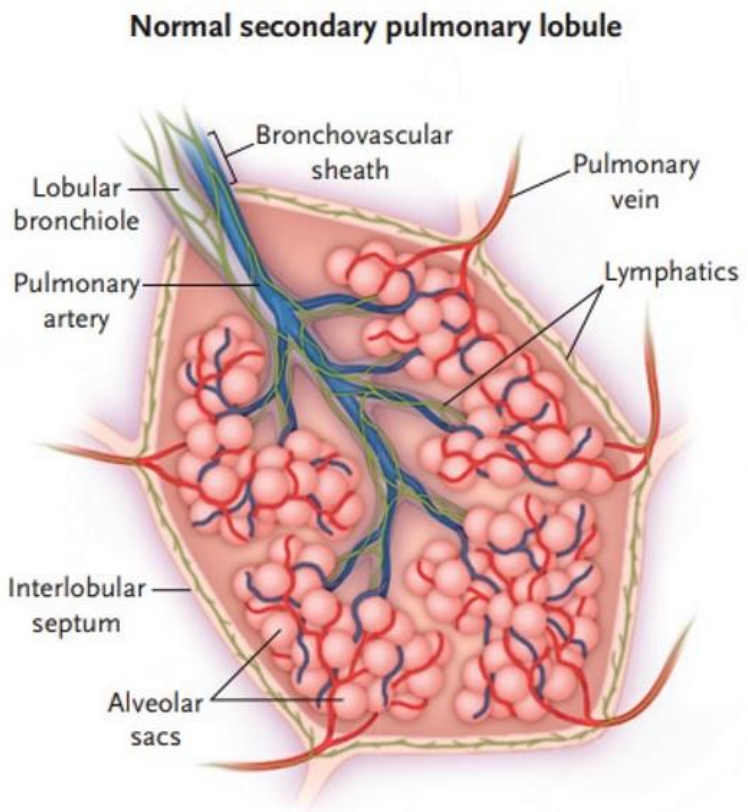
Identify the hilum



The Lung Parenchyma: Secondary Pulmonary Lobule






Normal secondary pulmonary lobule
Bronchovascular sheath
Lobular bronchiole
Pulmonary vein
Pulmonary artery
Lymphatics
Interlobular septum
Alveolar sacs



HIGH RESOLUTION CT (HRCT)

- **Define it:** The smallest unit of lung surrounded by connective tissue septa.
- **Components:**
  - **Centrilobular Artery & Bronchiole** (center)
  - **Lymphatics & Septa** (periphery)
  - **Lobular Core vs. Septal/Peripheral Disease:** This is the key to HRCT interpretation

- **Upper Abdomen** (Liver, Spleen, Adrenals - "Don't miss!")

Summary: Anatomy is the language of radiology. A systematic approach prevents errors.

Your Homework:

- Look at 1 normal CT this week (Radiopedia, normal CT chest case)
- Use an anatomy atlas

Thank you